RQ 1- Retrieval Pipeline

Recall @ K

Table S1. Recall at K comparison for the PURE dataset across Knowledge Graph based Retrieval and Vector based Semantic Retrieval pipelines.

Pipeline	K	Hits	Total_queries	Recall@K
KGR	1	36	36	1
KGR	2	36	36	1
KGR	3	36	36	1
KGR	5	36	36	1
KGR	15	36	36	1
KGR	20	36	36	1
VSR	1	35	40	0.875
VSR	2	39	40	0.975
VSR	3	40	40	1
VSR	5	40	40	1
VSR	15	40	40	1
VSR	20	40	40	1

This table shows recall values for the PURE dataset at different K settings. KGR achieves perfect recall from K equal to 1, while VSR reaches perfect recall at K equal to 3 and remains unchanged afterward. Since recall becomes stable after K equal to 3 for both pipelines, this value is selected as the retrieval depth for the PURE dataset.

Table S2. Recall at K comparison for the OpenCOSS dataset across KGR and VSR pipelines.

Pipeline	K	Hits	Total_queries	Recall@K
KGR	1	15	20	0.75
KGR	2	18	20	0.9

KGR	3	18	20	0.9
KGR	5	19	20	0.95
KGR	10	20	20	1
KGR	15	20	20	1
KGR	20	20	20	1
VSR	1	17	20	0.85
VSR	2	20	20	1
VSR	3	20	20	1
VSR	5	20	20	1
VSR	10	20	20	1
VSR	15	20	20	1
VSR	20	20	20	1

This table reports recall values for the OpenCOSS dataset across different K settings. VSR reaches perfect recall at K equal to 2 and remains stable, while KGR shows gradual improvement and approaches stable recall around K equal to 5, with only minor changes afterward. Based on this saturation trend, K equal to 5 is selected as the retrieval depth for the OpenCOSS dataset to maintain consistency while avoiding unnecessary expansion of the candidate set.

Table S3. Recall at K comparison for the WorldVista dataset across KGR and VSR pipelines.

Pipeline	K	Hits	Total_queries	Recall@K
KGR	1	60	69	0.869565
KGR	2	63	69	0.913043
KGR	3	66	69	0.956522
KGR	5	66	69	0.956522
KGR	10	67	69	0.971014
KGR	15	67	69	0.971014
KGR	20	67	69	0.971014
VSR	1	63	70	0.9

VSR	2	66	70	0.942857
VSR	3	67	70	0.957143
VSR	5	68	70	0.971429
VSR	10	69	70	0.985714
VSR	15	69	70	0.985714
VSR	20	70	70	1

This table presents recall values for the WorldVista dataset across different K settings. Both KGR and VSR show gradual improvement with increasing K, with recall becoming relatively stable around K equal to 5 and only minor gains observed afterward. Based on this trend, K equal to 5 is selected as the retrieval depth for the WorldVista dataset, as it captures most relevant candidates while avoiding unnecessary increase in retrieval size.

Table S4. Recall at K comparison for the UAV dataset across KGR and VSR pipelines.

Pipeline	K	Hits	Total_queries	Recall@K
KGR	1	32	34	0.941176
KGR	2	32	34	0.941176
KGR	3	32	34	0.941176
KGR	5	32	34	0.941176
KGR	10	32	34	0.941176
KGR	15	32	34	0.941176
KGR	20	32	34	0.941176
VSR	1	33	35	0.942857
VSR	2	33	35	0.942857
VSR	3	33	35	0.942857
VSR	5	33	35	0.942857
VSR	10	33	35	0.942857
VSR	15	33	35	0.942857

VSR	20	33	35	0.942857

This table shows recall values for the UAV dataset across different K settings. Both KGR and VSR exhibit stable recall from the early K values, with no observable improvement beyond K equal to 2. Since recall remains unchanged for larger K values, K equal to 2 is selected as the retrieval depth for the UAV dataset to maintain efficiency while preserving retrieval effectiveness.

Table S5. Recall at K comparison for the IBM UAV dataset across KGR and VSR pipelines.

Pipeline	K	Hits	Total_queries	Recall@K
KGR	1	405	651	0.62212
KGR	2	494	651	0.758833
KGR	3	517	651	0.794163
KGR	5	548	651	0.841782
KGR	10	595	651	0.913978
KGR	15	618	651	0.949309
KGR	20	633	651	0.97235
VSR	1	773	788	0.980964
VSR	2	777	788	0.986041
VSR	3	777	788	0.986041
VSR	5	780	788	0.989848
VSR	10	783	788	0.993655
VSR	15	783	788	0.993655
VSR	20	784	788	0.994924

This table reports recall values for the IBM UAV dataset across different K settings. Unlike smaller datasets, both KGR and VSR continue to show noticeable improvements with increasing K, indicating the need for deeper retrieval to capture relevant candidates. Since recall does not saturate at lower K values and reaches its highest level at K equal to 20, this value is selected as the retrieval depth for the IBM UAV dataset to ensure comprehensive retrieval coverage.

Table S6. Cross dataset summary of retrieval performance and sustainability metrics for KGR and VSR pipelines.

Dataset	Pipeline	Top - K	Recall @ K	Total_Energy_kWh	Total_Carbon_kgCO2e	Avg_Energy_per_Retrieval_kWh	Avg_Carbon_per_Retrieval_kgCO2e	Total_Latency_s
open_coss	KGR	5	0.95	0.000003	0.000002	0.00000001	0.00000001	979.592705
open_coss	VSR	5	1	0.000012	0.000007	0.00000005	0.00000003	664.638795
pure	KGR	3	1	0.000002	0.000001	0.00000001	0.00000001	555.493388
pure	VSR	3	1	0.000008	0.000005	0.00000006	0.00000004	380.060194
uav	KGR	2	0.94	0.000003	0.000002	0.00000001	0.00000001	965.351021
uav	VSR	2	0.94	0.000011	0.000007	0.00000005	0.00000003	657.999271
world_vista	KGR	5	0.95	0.000004	0.000002	0.00000001	0.00000001	1242.403111
world_vista	VSR	5	0.97	0.000014	0.000009	0.00000005	0.00000003	841.063754
IBM_UAV	KGR	20	0.97	0.001026	0.000644	0.00000008	0.00000005	37582.19716
IBM_UAV	VSR	20	0.99	0.000869	0.000546	0.00000006	0.00000004	40462.22191

This table provides an overall comparison of KGR and VSR across all datasets using the selected K values. It summarizes retrieval effectiveness along with energy consumption, carbon emissions, and latency. The results show that KGR generally achieves comparable recall with lower energy and carbon usage for small and medium datasets, while VSR demonstrates slightly higher recall and competitive efficiency in the large IBM UAV dataset.

RQ 2 - Model Ranking

Table S7. Model level ranking based on performance and sustainability metrics across all datasets.

model	f1_macro	mean_latency	mean_energy	total_carbon	eco_score	rank_f1	rank_latency	rank_eco
DeepSeek	0.295637669	2.172205324	0.000143899	0.445723311	0.663276211	3	2	2
Llama	0.370637085	2.362363859	0.000157004	0.783091794	0.473299666	2	3	3
Mistral	**0.538997426**	**1.042146803**	**7.25E-05**	**0.362668554**	**1.486198404**	**1**	**1**	**1**

This table presents a comparative ranking of DeepSeek, Llama, and Mistral using Macro F1, latency, energy consumption, total carbon, and Eco Score. Mistral achieves the highest Macro F1 while also showing the lowest latency and energy usage, resulting in the best Eco Score and overall ranking. Llama demonstrates moderate performance but incurs higher latency and carbon impact, whereas DeepSeek shows lower classification performance with relatively balanced latency and sustainability metrics. Overall, the results indicate that Mistral provides the most favorable balance between predictive effectiveness and environmental efficiency.

RQ 2 - Dataset Level

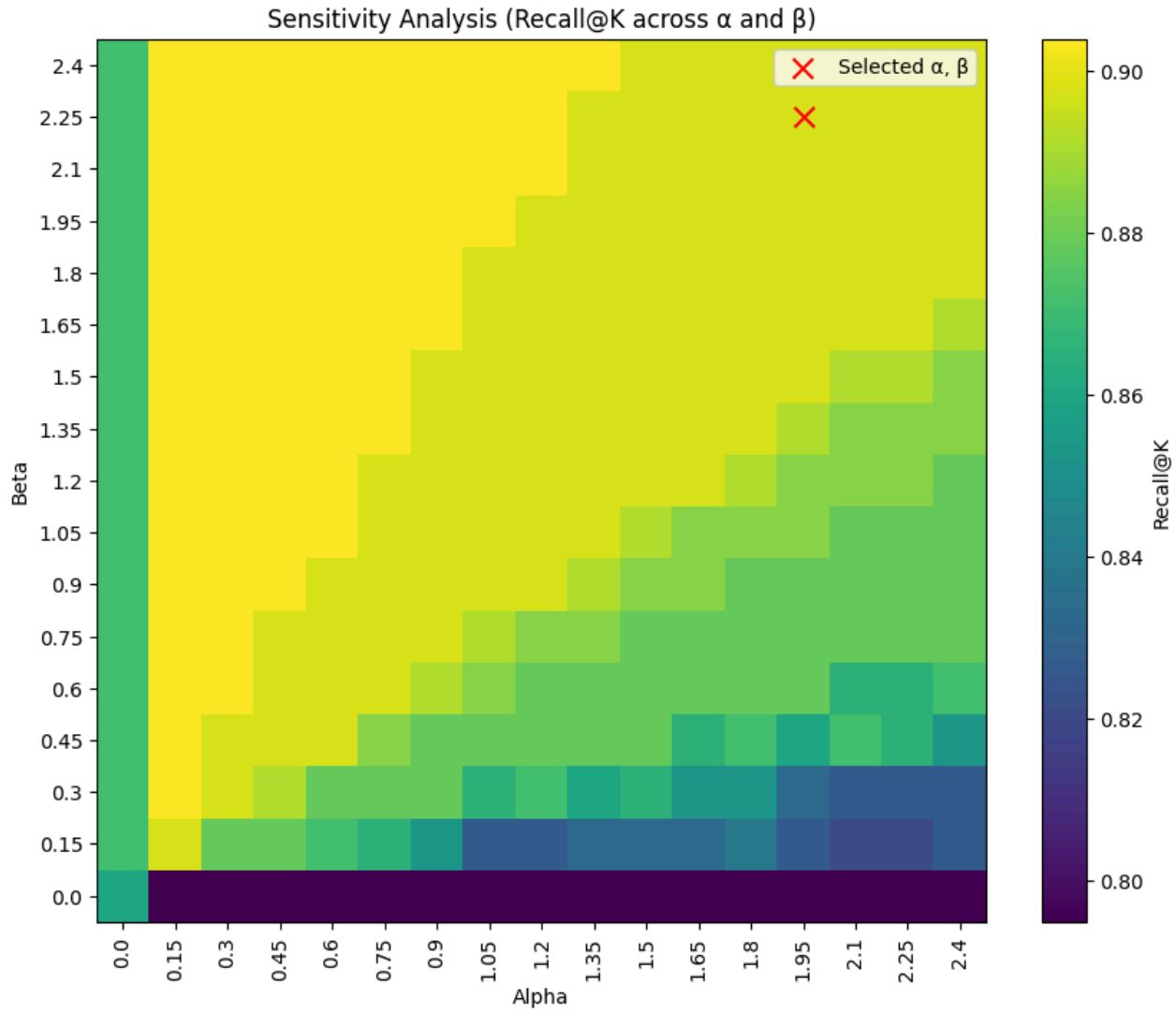
Table S8. Dataset level comparison of SLM performance and carbon emissions across DeepSeek, Llama, and Mistral.

Dataset	SLM Model	f1_macro	precision_macro	recall_macro	Total Carbon emission
IBM_UAV_CDN	DeepSeek	0.237532035	0.266688922	0.302399344	0.095319104
	Llama	0.381831645	0.425493531	0.483741403	0.099794066
	Mistral	0.280441594	0.432809523	0.349850554	0.045678414

Open_Coss	DeepSeek	0.267418637	0.465454958	0.460074059	0.120815762
	Llama	0.245787985	0.491676032	0.483593001	0.238932318
	Mistral	0.586193718	0.626581199	0.596981762	0.11519401
PURE	DeepSeek	0.344567625	0.472091942	0.45500322	0.038247391
	Llama	0.356374547	0.493997599	0.495326171	0.072756047
	Mistral	0.502457715	0.629154055	0.522937258	0.03280992
UAV	DeepSeek	0.376918103	0.488458676	0.507015306	0.039222548
	Llama	0.489261764	0.572297183	0.594120728	0.075706833
	Mistral	0.688108169	0.793883328	0.666801477	0.035503478
World_Vista	DeepSeek	0.251751945	0.355568292	0.351067145	0.152118507
	Llama	0.379929483	0.550922018	0.565324143	0.295902531
	Mistral	0.637785933	0.664694739	0.632315411	0.133482731

This table summarizes Macro F1, precision, recall, and carbon emissions for each model across datasets. Mistral emerges as the best performing model on most datasets, achieving the highest Macro F1 on OpenCoss (0.586), PURE (0.502), UAV (0.688), and World Vista (0.638), while also maintaining comparatively lower carbon emissions. In contrast, Llama performs best on the IBM UAV dataset with a Macro F1 of 0.3818, where Mistral records a lower value of 0.280. This is the only dataset where Mistral underperforms, which can be attributed to the presence of duplicate requirements and synthetic conflict pairs that introduce additional ambiguity during classification. Overall, the results indicate that Mistral provides the most consistent performance across datasets, with the IBM UAV dataset representing a distinct case influenced by its data composition.

Sensitivity Analysis for RQ 2:



A sensitivity analysis was conducted to determine the optimal values of the weighting parameters alpha, beta and gamma with respect to the performance metric Recall@K. The heatmap illustrates the variation in Recall@K across different combinations of these parameters. From the analysis, it can be observed that several combinations yield comparable high performance; however, selecting excessively large values for both parameters leads to increased **computational overhead** without significant improvement in **retrieval performance**. Therefore, the configuration alpha = 2.1 and beta = 2.25 was selected as it provides near-optimal Recall@K while maintaining computational efficiency. This choice ensures that the model remains computationally inexpensive, which in turn reduces processing time and energy consumption during large-scale trajectory evaluation. Additionally, the parameter gamma is not included in

the heatmap visualization because preliminary experiments indicated that variations in gamma produced negligible influence on the trajectory ranking results. Consequently, gamma was fixed to a constant value in the experiments to simplify the sensitivity analysis and focus on the parameters that most significantly impact performance.

Statistical Significance:

We conducted statistical significance testing using McNemar's test to compare the retrieval effectiveness of KAG and RAG across all datasets at their respective K values. For PURE, OPEN_COSS, UAV, and WORLD_VISTA, both pipelines achieved nearly identical Recall@K scores, with negligible disagreement between their predictions and non-significant p-values ($p = 1.0$), indicating no statistically meaningful difference in performance. In contrast, for the IBM_UAV dataset, RAG significantly outperformed KAG ($p < 0.001$) with a large effect size, demonstrating a clear advantage of embedding-based retrieval in this setting. This difference can be attributed to the structural complexity and high entity frequency in IBM_UAV also due to a few requirements being , where graph-based traversal introduces noise due to densely connected entities, whereas semantic embeddings provide better discrimination. It is also important to note that certain requirements in the IBM_UAV dataset contain noise and special characters, which adversely affect the performance of the graph-based (KAG) retrieval pipeline. Since KAG relies on structured entity extraction and exact matching within the knowledge graph, such irregularities can lead to incomplete or incorrect entity mapping, thereby reducing retrieval effectiveness. In contrast, embedding-based retrieval (RAG) is more robust to such noise due to its semantic representation. We intentionally avoid aggressive preprocessing or text normalization to remove these artifacts, as doing so would introduce additional computational overhead and energy consumption, which is not required for the other datasets. This design choice ensures consistency in evaluation and reflects a realistic deployment scenario, where preprocessing pipelines are often constrained by efficiency considerations.

Overall, these results suggest that while graph-based retrieval is sufficient for moderately structured datasets, embedding-based retrieval offers greater robustness in complex requirement spaces.

Dataset	K	KAG_Recall	RAG_Recall	b(KAG>RAG)	c(RAG>KAG)	p_value	effect_size_g	anchors_compared
PURE	3	0.97222	0.9722	1	1	1.0	0.0	36
OPEN_COSS	5	1.0	1.0	0	0	1.0	0.0	20
UAV	2	0.9412	0.9412	0	0	1.0	0.0	34
WORLD_VISTA	5	0.95652	0.9714	2	3	1.0	0.2	69
IBM_UAV	20	0.9677	0.9946	4	21	0.00091	0.68	651